Mustafa Sameen

Ph.D. Student in Civil and Coastal Engineering
SERMoS Lab, University of Florida
Gainesville, Florida, USA

mustafasameen@ufl.edu
Homepage, Google Scholar, GitHub
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RESEARCH AREAS

AI agents: LLM agents and multi-agent simulations of household decision-making.

Human mobility: behavioral modeling and benchmarks grounded in individual GPS records.

Disaster mobility: travel before hazards, evacuation, and post-disaster recovery.

EDUCATION

2025-present University of Florida, Gainesville, FL
Ph.D. in Civil and Coastal Engineering. Advisor: Xilei Zhao.

2025-present University of Florida, Gainesville, FL
M.S. in Artificial Intelligence Systems.

2021-25 Colorado College, Colorado Springs, CO
B.A. in Computer Science and Mathematics, *magna cum laude*.
Study abroad: Aquincum Institute of Technology, Budapest.

RESEARCH EXPERIENCE

2025-present University of Florida, SERMoS Lab
Graduate Research Assistant (Jul 2025 - present)
Develop LLM agents that simulate household negotiation (PEMAND) and use behavioral theory to model ridesourcing choices (SAPA).
Test whether LLMs can recover mobility descriptors from GPS records (GAUGE).
Study post-earthquake mobility in Noto and wildfire evacuation using GPS data.

2025 MIT JTL-Transit Lab
Summer Research Intern (Jun - Jul 2025)
Combined policy documents with GTFS and GIS data in a spatial retrieval pipeline for the Chicago Transit Authority. Prototyped LLM agents for reasoning about transit delays and rerouting. Advisors: Haris Koutsopoulos and Muhammad Usama.

RESEARCH SOFTWARE

GAUGE Benchmark and code for testing what LLMs can infer from mobility records.

GHOST Python package for inferring home locations from GPS data using grid-based stay time.

EMBER Python package for inferring wildfire evacuation behavior from GPS-derived mobility data.

PUBLICATIONS

Conference papers

2026 **Sameen, M.**, Vibhute, M., and Zhao, X. "Look Both Ways: Reconstructing Two Hidden Months of Post-Earthquake Mobility." *HuMob workshop, ACM SIGSPATIAL*. 10.1145/3849738.3858661

2026 **Sameen, M.** "GAUGE: How Large Language Models Recover Mobility Descriptors from Individual Records." *UMFM workshop, ACM SIGSPATIAL*. 10.1145/3849729.3856126

2026 Jiang, S., **Sameen, M.**, Recalde, A., and Zhao, X. "GeoAI-Act: GeoAI-based Activity Inference from GPS Trajectories." *GeoAI workshop, ACM SIGSPATIAL*. 10.1145/3849726.3856090

2026 Recalde, A., **Sameen, M.**, Zhang, X., and Zhao, X. "Design and Validation of a Grid-Based Home Detection via Stay-Time (GHOST) Software for Mobile Location Data." *Transportation Research Board 105th Annual Meeting*. arXiv:2605.20429

Preprints

2026 Sun, Y.[*], **Sameen, M.**[*], Zhang, Y., Gu, R., Vibhute, M., Wu, C., Lei, Y., and Zhao, X. "PEMAND: Persona-Enriched Multi-Agent Negotiation for Household Decision-Making." *arXiv preprint*. arXiv:2604.10475

2025 **Sameen, M.**, Zhang, X., and Zhao, X. "Synthesizing Attitudes, Predicting Actions (SAPA): Behavioral Theory-Guided LLMs for Ridesourcing Mode Choice Modeling." *arXiv preprint*. arXiv:2509.18181

[*]Equal contribution (co-first authors).

AWARDS AND HONORS

2026 4th place out of 80 teams, HuMob Challenge, ACM SIGSPATIAL.

2026 ACM SIGSPATIAL Travel Grant.

2025 Phi Beta Kappa, Colorado College.

2025 Euclid Scholar, Colorado College.

2025 Thomas Post Rawles Mathematics Prize, upper division, Colorado College.

2022 Thomas Post Rawles Mathematics Prize, lower division, Colorado College.

TEACHING

EGN 5215 **Machine Learning Applications in Civil Engineering**
Teaching Assistant, University of Florida. Instructor: Xilei Zhao.